Programming for AI

Quiz 1

Question 1

Write a function **process_coordinates(points, min_dist)** that takes a list of strings representing 2D coordinates.

1. Each string contains a PointID and an (x, y) coordinate pair separated by colons (e.g., `"P1:3:4"`).
2. Define a "distance" rule that calculates the Euclidean distance from the origin.
3. Use a List Comprehension to apply this lambda, but only include points whose distance is strictly greater than `min_dist`.
4. Return a list of dictionaries with keys `pointId` and `distance`. The final list must be **sorted by distance** in descending order.

Example:

```
["P1:3:4", "P2:1:1", "P3:0:5", "P4:10:10", "P5:2:2"]
```

Question 2

Write a function **analyze_inventory(stock_data, min_value)** that takes a list of strings representing warehouse stock.

- Each string contains a SKU, Category, UnitPrice, and StockCount separated by colons (e.g., `"SKU45:Electronics:200.0:15"`).
- Define a "valuation" rule that calculates the total value of that stock item (UnitPrice × StockCount) using lambda function.
- Use a List Comprehension to apply this logic, but only include items where the total valuation is greater than `min_value`.
- Return a list of dictionaries with keys `sku`, `category`, and `totalValue`. The final list must be **sorted alphabetically by category.**

Example

```
["A1:Office:10:50","B2:Hardware:100:2","C3:Office:5:10","D4:Hardware:50:20"
, "D5:Accounts:20:80"]
```